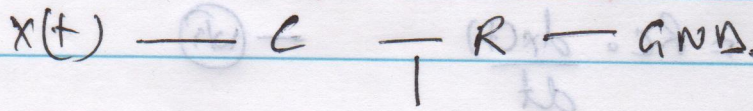


$$C = 0.01 \mu$$

High pass.



Current across capacitor, $i(t)$

$$i(t) = C \frac{dV}{dt}$$

$$= C \cdot \frac{dx(t) - y(t)}{dt} \quad \text{--- (i)}$$

Voltage Across Resistor.

$$V_R = i(t) \cdot R$$

$$= C \cdot \frac{dx(t) - y(t)}{dt} \cdot R$$

$$= RC \cdot \frac{dx(t) - y(t)}{dt}$$

$$\Rightarrow V_R = RC \left[\frac{dx(t)}{dt} - \frac{dy(t)}{dt} \right] \quad \text{--- (ii)}$$

$$\Rightarrow y(t) = RC \frac{dx(t)}{dt} - RC \frac{dy(t)}{dt} \quad \text{--- (iii)}$$

$$\Rightarrow y(t) + RC \frac{dy(t)}{dt} = RC \frac{dx(t)}{dt} \quad \text{--- (iv)}$$

$$\Rightarrow a_0 y(t) + a_1 \frac{dy(t)}{dt} = b_0 \frac{dx(t)}{dt} \quad \text{--- (v)}$$

$\Rightarrow$ where $a = 1$, $a_1 = RC$, $b_0 = 1$, $b_1 = 0$, all the coefficients are real.

$\Rightarrow$ Note term $\frac{dx(t)}{dt}$ appears in system when input reacts

with dynamic [reactive] element like capacitor.

① Convert the differential equation to Laplace transform.

$$y(t) + RC \frac{dy(t)}{dt} = RC \cdot \frac{dx(t)}{dt} \quad \text{--- (vi)}$$

Assuming zero initial conditions

$$\text{i.e. } x(0) = y(0) = 0.$$

Using Rules.

$$\rightarrow \mathcal{L}\{y(t)\} = Y(s).$$

$$\rightarrow \mathcal{L}\left\{\frac{dy(t)}{dt}\right\} = s \cdot Y(s) - (t) \cdot y(0) =$$

$$\rightarrow \mathcal{L}\left\{\frac{dx(t)}{dt}\right\} = s \cdot X(s)$$

Applying Laplace transformation in eqn (vi)

$$Y(s) + RC s \cdot Y(s) = RC s \cdot X(s)$$

Solve for $Y(s)$

$$Y(s) = \frac{RC s \cdot X(s)}{1 + RC s}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{RC s}{1 + RC s} \quad \text{--- (vii)}$$

First order High pass filter.
Laplace domain transfer function.

$$H(s) = \frac{T s}{1 + T s} \quad \text{--- (viii) where } T = RC$$

Now

$$\lim_{s \rightarrow \infty} H(s) = 1$$

$$H(0) = \frac{0}{1} = 0 \quad \text{Blocks Low Freq.}$$

HPF

$$\omega^2 \quad (-1)^2 T^2$$

Convert to Frequency domain/response. ~~HPF~~ $H(s) \rightarrow H(j\omega)$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{RCs}{1+RCs} \quad \text{--- (ix)}$$

Substitute s with $j\omega$.

$$H(j\omega) = \frac{RCj\omega}{1+RCj\omega} = \frac{Tj\omega}{Tj\omega+1}$$

$$\text{Magnitude } |H(j\omega)| = \left| \frac{j\omega T}{1+Tj\omega} \right| = \frac{\omega T}{\sqrt{1+\omega^2 T^2}}$$

$$|H(j\omega)| = \frac{\omega T}{\sqrt{1+\omega^2 T^2}}$$

Phase $\angle H(j\omega) = \angle j\omega T - \angle (1+Tj\omega) \rightarrow \text{From (ix)}$
 $\rightarrow \angle j\omega T \rightarrow$ purely imaginary & positive
 $\rightarrow 1+Tj\omega \rightarrow \tan^{-1}\left(\frac{\omega T}{1}\right)$

$$\angle H(j\omega) = 90^\circ - \tan^{-1}(\omega T) \quad \text{phase}$$

$$\angle H(j\omega) = \frac{\pi}{2} - \tan^{-1}(\omega T)$$

when $\omega \rightarrow 0^\circ$

$$|H(j\omega)| = \frac{0T}{\sqrt{1+0^2 T^2}} = 0$$

$$\angle H(j\omega) = 90^\circ - \tan^{-1}(0 \cdot T) = 90^\circ$$

when $\omega \rightarrow \infty$

$$|H(j\omega)| = \frac{\infty T}{\sqrt{1+\infty^2 T^2}} = 1$$

$$\angle H(j\omega) = 90^\circ - \tan^{-1}(\infty \cdot T) \Rightarrow 90^\circ - 90^\circ = 0^\circ$$

HPF

$$|H(j\omega)| = \frac{\omega z}{\sqrt{1 + \omega^2 z^2}} ; \angle H(j\omega) = 90^\circ - \tan^{-1}(\omega z)$$

Summary.

	$\omega \rightarrow 0$	$\omega \rightarrow \infty$
$ H(j\omega) $	0	1
$\angle H(j\omega)$	90°	0°
Gain _{dB}	$20 \log_{10}(H(j\omega))$	

② Convert from s plane to z Domain. ②

Common method.

Method

$s \rightarrow$ Substitution.

Notes.

Forward Euler

$$s \approx \frac{z-1}{T}$$

Simple, less accurate.

Backward Euler

$$s \approx \frac{1-z^{-1}}{T}$$

More stable, implicit method

Bilinear Transform.

$$s \approx \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

Warps frequency, widely used

Bilinear Transform: Maps entire $j\omega$ axis into unit circle.
in z plane & preserve stability.

$$s = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

$$\frac{(1-s-1)T}{(1-s+1)T}$$

$$(1-s-1)T = (1-s+1)T$$

$$(1-s-1)T = (1-s+1)T$$

$$1 + \frac{(1-s-1)T}{(1-s+1)T}$$

(*) Convert ~~S~~ First Order HPF to z domain.

Using Bilinear Transform,

$$s = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}} \quad \text{where } T = \frac{1}{F_s}$$

$$\text{First order } H(s) = \frac{Y(s)}{X(s)} = \frac{KCs}{1+KCs} = \frac{Zs}{1+Zs}$$

substituting s

$$Z \cdot \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}$$

$$1 + \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}} \cdot Z$$

$$= \frac{Z \cdot 2(1-z^{-1})}{T(1+z^{-1})} = \frac{2Z(1-z^{-1})}{T(1+z^{-1}) + 2Z(1-z^{-1})}$$

$$= \frac{2Z(1-z^{-1})}{T + Tz^{-1} + 2Z - 2Zz^{-1}} = \frac{2Z(1-z^{-1})}{z^{-1}(T - 2Z) + (T + 2Z)}$$

Normalizing.

Dividing Numerator & Denominator by $(T + 2Z)$

$$\frac{2Z(1-z^{-1})}{T + 2Z} \Rightarrow \frac{2Z}{T + 2Z} (1-z^{-1})$$

$$\frac{z^{-1}(T - 2Z)}{T + 2Z} + \frac{T + 2Z}{T + 2Z} = \frac{z^{-1} \left(\frac{T - 2Z}{T + 2Z} \right) + 1}{1}$$

Solving/continuing.

$$\alpha = \frac{2Z/T}{1 + \frac{2Z}{T}}$$

$$\beta = \frac{1 - \frac{2Z}{T}}{1 + \frac{2Z}{T}}$$

$$1 = \frac{1 - 2Z/T}{1 + 2Z/T}$$

$$\text{if } \alpha = \frac{2Z}{T + 2Z}$$

$$\beta = \left(\frac{T - 2Z}{T + 2Z} \right)$$

$$\Rightarrow \frac{\alpha(1 - z^{-1})}{z^{-1}\beta + 1} = H(z) = \frac{Y(z)}{X(z)}$$

$$\Rightarrow \alpha(1 - z^{-1}) \cdot X(z) = Y(z) \cdot z^{-1}\beta + Y(z)$$

Taking Inv $\Rightarrow$ Transformations.

$$z^{-1} Y(z) \Leftrightarrow y[n-1]$$

$$Y(z) = y[n]$$

$$z^{-1} X(z) \Leftrightarrow x[n-1]$$

$$X(z) = x[n]$$

$$\Rightarrow \alpha(X(z) - X(z)z^{-1}) = Y(z) - z^{-1}\beta + Y(z)$$

$$\Rightarrow \alpha(x[n] - x[n-1]) = \beta y[n-1] + y[n]$$

$$\Rightarrow y[n] = -\beta y[n-1] - \alpha x[n-1] + \alpha x[n]$$

To solve for eqn. we need T_s = which is $\frac{1}{f_s}$
and $RC = Z$ constant, for solving α & β .

as

$$\alpha = \frac{2Z/T}{1 + 2Z/T}$$

$$\beta = \frac{1 - 2Z/T}{1 + 2Z/T}$$

$$\text{if } K = \frac{2Z}{T} = \frac{2RC}{T_s} \Rightarrow \alpha = \frac{K}{1 + K}$$

$$\beta = \frac{1 - K}{1 + K}$$

Compute a HPF with -3dB at 10kHz.

~~area~~ $f_c = 10 \text{E}03$

$$f_c = \frac{1}{2\pi RC} \Rightarrow RC = \frac{1}{10 \text{E}03 \times 2 \times \pi} = \frac{1}{62831.85 \text{E}03} \Rightarrow Z = 15.91 \text{E}05$$

General form of HPF First order.

$$y[n] = -\beta y[n-1] - \alpha x[n-1] + \alpha x[n]$$

where

$$\alpha = \frac{2Z/T}{1 + 2Z/T}$$

$$\beta = \frac{1 - 2Z/T}{1 + 2Z/T}$$

If $K = \frac{2Z}{T} \Rightarrow \alpha = \frac{K}{1+K}$

$$\beta = \frac{1-K}{1+K}$$

If $F_s = 2 \times f_c \Rightarrow 20 \text{kHz}$

$\Rightarrow$ Making it 25K see the effect

[20 @ 100] kHz.

Solve for K.

~~area~~ $K = \frac{2 \times 6.2857 \text{E}4}{20 \text{E}03} = \frac{6.2857 \times 2 \times 15.91 \text{E}05}{20 \text{E}03}$

$$\alpha = \frac{K}{1+K}$$

$$\beta = \frac{1-K}{1+K}$$

$$\frac{1.8915 \text{E}09}{1 + 1.8915 \text{E}09}$$

$$\beta = \frac{1-K}{1+K}$$

$$= \frac{2 \times 1}{20 \text{E}03} \times 15.91 \text{E}05$$

$$\beta = 0.220$$

$$1 + \frac{2 \times 1}{20 \text{E}03} \times 15.91 \text{E}05$$

$$\alpha = 0.3890$$

$$\therefore y[n] = -0.220 y[n-1] - 0.3890 x[n-1] + x[n]$$